5.4 Problems

(1) Define a relation R on $\mathbb{R}$ by saying that xRy if $|x - y| < 1$. Is R an equivalence relation? Which of the properties of a relation are satisfied by R?

(2) Is the relation R on $\mathbb{R}$, given by xRy if $x - y \in \mathbb{Z}$, an equivalence relation? What about the relation S given by xSy if $x - y \in \mathbb{Q}$?

(3) In Definition 5.5, it seems as though reflexivity should follow from symmetry and transitivity: $a \sim b$ implies $b \sim a$, so we have $a \sim b$ and $b \sim a$ and thus $a \sim a$. Why is this wrong? What is the flaw in the argument? Also, give an example of a binary relation on a set X that satisfies symmetry and transitivity but not reflexivity.

(4) Prove that the two examples of equivalence relations (similarity of triangles and equivalence of function inputs) given right after Definition 5.5 are actually equivalence relations, by checking that they satisfy the three properties in the definition.

(5) Suppose $\sim$ and $\equiv$ are two equivalence relations on a set X. Define a new relation R on X by saying that xRy if either $x \sim y$ or $x \equiv y$. Is R necessarily an equivalence relation? Prove or find a counterexample.

(6) Suppose $\sim$ and $\equiv$ are two equivalence relations on a set X. Define a new relation R on X by saying that xRy if both $x \sim y$ and $x \equiv y$. Is R necessarily an equivalence relation? Prove or find a counterexample.

(7) Suppose $\sim$ is an equivalence relation on a finite set X with n elements. Suppose, furthermore, that every equivalence class has exactly k elements. How many equivalence classes are there?

(8) Suppose that R is a reflexive and symmetric relation on a set X. Define a new relation $\sim$ on X by saying that $x \sim y$ if there exists a positive integer n and elements $x_1, x_2, x_3, \ldots, x_n \in X$ such that $x_1 = x$, $x_n = y$, and for $1 \leq i \leq n-1$, we have $x_i R x_{i+1}$. Prove that $\sim$ is an equivalence relation on X.

(9) Let $X = \mathbb{Z} \times (\mathbb{Z} \backslash \{0\})$ be the set of pairs of integers, where the second one cannot be zero. Define a relation $\sim$ on X by declaring that $(a, b) \sim (c, d)$ if $ad = bc$. Prove that $\sim$ is an equivalence relation. Explain why the equivalence classes are related to the rational numbers $\mathbb{Q}$.

*Equivalence Relation

10/20/26

T+P Week 5: Equivalence Relations

Alexander

5.1: √Reflexivity: $|x-x|<1$ is just $0<1$, so True.

√Symmetry: $|x-y|=|y-x|$, so if $|x-y|<1$ then so is $|y-x|$. True.

X Transitivity: $|x-y|<1$, $|y-z|<1$, is $|x-z|<1$?

Add the first 2 to get $|x-y+y-z|=|x-z|<2$, so not necessarily true. Thus, $|x-y|<1$ is NOT an ER*.

5.2: Reflexivity: $x-x=0\in\mathbb{Z}$ (Z) True.

Symmetry: $x-y\in\mathbb{Z}$, $y-x=-(x-y)\in\mathbb{Z}$. True.

Transitivity: $x-y\in\mathbb{Z}$, $y-z\in\mathbb{Z}$, does $(x-z)\in\mathbb{Z}$?

$(x-y)+(y-z)=(x-z)$, and adding two integers must result in another integer. True.

Thus, $x-y\in\mathbb{Z}$ for some xRy is an ER.

Now for $\mathbb{Q}$. $(x-y\in\mathbb{Q})$

Reflexivity: $x-x=0\in\mathbb{Q}$. True.

Symmetry: Wait... $\mathbb{Z}\in\mathbb{Q}$, so $\mathbb{Q}$ ($x-y$ in $\mathbb{Q}$) is an ER!

But while $\mathbb{Z}$ has one ER in the form $x-y$, $\mathbb{Q}$ has infinitely many due to the different "types" of denominators for each $\mathbb{Z}$.

5.3: Recall that equivalence is that $a\sim a$ for all $\underline{a\in X}$. This only has $\underline{\text{one}}$ $a\in X$ such that $a\sim a$. Example: $X=\{\emptyset\}$, R is (a,a). If (a,a), then (a,a), but not (a,a) for all a.

†Equivalence Classes
*sorry, used a and b instead of x and y.

5.4: Reflexivity: $\triangle ABC \sim \triangle ABC$. Yes, they're congruent too.
Symmetry: $\triangle ABC \sim \triangle DEF$, so $\triangle DEF \sim \triangle ABC$. ($\triangle DEF$ and $\triangle ABC$ have same angles.)
Transitivity: $\triangle ABC \sim \triangle DEF$, $\triangle DEF \sim \triangle GHI$, then $\triangle ABC$ must also be similar to $\triangle GHI$ by both factors multiplied.
So similarity IS an ER.

Reflexivity: $f(x_1)=f(x_1)$. Uhh... yes! So True. (They're the same.)
Symmetry: $f(x_1)=f(x_2)$, $f(x_2)=f(x_1)$. Just like $=$, True.
Transitivity: $f(x_1)=f(x_2)$, $f(x_2)=f(x_3)$, $f(x_1)=f(x_3)$ by equality. (If $a=b$, $b=c$, then $a=c$)
So $f(x_1)=f(x_2)$ as $x_1 \sim x_2$ IS an ER.

5.5: Reflexivity: Either $a \sim a^*$ or $a \equiv a$, both are True.
Symmetry: Because $a \sim b$ (or $a \equiv b$), $b \sim a$ (or $b \equiv a$).
Transitivity: If $a \sim b$ (Handpicked, but still) and $b \equiv C$... Uhoh.
That's not necessarily true, so aRb is NOT an ER.

5.6: Reflexivity: $a \sim a$ AND $a \equiv a$, which is True.
Symmetry: $a \sim b$ so $b \sim a$ AND $a \equiv b$ so $b \equiv a$. True.
Transitivity: $a \sim b$ and $b \sim c$, so $a \sim c$ AND same for $\equiv$. True.
Here, aRb IS an ER.

5.7: If all EC† are k long, then k must be a factor of the number of elements in X (it divides evenly.) Thus, there are just $\frac{|X|}{k}$ ECs.

5.8: Reflexivity: $x \sim x$, so if x_1 and x_n are both x, then it's just $n=2$ and $x_1 R x_2$. True.
Symmetry: $x \sim y$, so $y \sim x$. Because R is symmetric, we can trace back from x_n to x_1 (or y to x). True.
Transitivity: It's not given in R, so we need a Proof. $x \sim y$ and $y \sim z$, so $x_1 R x_2 R \ldots R x_n$ and $x_n R x_{n+1} R \ldots R x_m$. We can stitch these together because they share x_n, so ultimately $x_1 R \ldots R x_m$ and $x \sim z$. True.
Thus, $\sim$ Is an ER.

5.9: Reflexivity: $(a,b) \sim (a,b)$, so $ab=ba$, so True.
Symmetry: If $ad=bc$, then $bc=ad$. This is = True by being an ER.
Transitivity: $(a,b) \sim (c,d)$, $\sim (e,f)$, so $ad=bc$ and $cf=de$. Show that $af=be$. $adcf=bcde$, $\div cd$ both sides, $af=be$. Thus, $(a,b) \sim (c,d)$ Is an ER.
Note that $ad=bc$ means $\frac{a}{b}=\frac{c}{d}$. This is the definition of $\mathbb{Q}$, rationals, or two integers in a fraction. ECs are all $\mathbb{Q}$ that reduce to an $\frac{a}{b}$. ■ (QED)